

Terraform Modules: Root Module → Child Module → Variables

Core idea: **the root module does not automatically share its variables with the child.** The root explicitly passes values through the **module block**; the child declares matching **variable blocks** to receive them.

1 ROOT MODULE

variables.tf

Defines inputs accepted by the root configuration.

```
variable "vpc_cidr" {
  type = string
}
variable "vpc_name" {
  type = string
}
variable "environment" {
  type = string
}
```

main.tf

Calls the reusable VPC child module and passes values.

```
module "vpc" {
  source = "../modules/vpc"
  vpc_name = var.vpc_name
  cidr = var.vpc_cidr
  env = var.environment
}
```

2 CHILD MODULE: VPC

variables.tf

These are the child's input variables. They receive values passed by the root.

```
variable "vpc_name" {
  type = string
}
variable "cidr" {
  type = string
}
variable "env" {
  type = string
}
```

main.tf

Uses received inputs as **var.<name>**.

```
resource "aws_vpc" "this" {
  cidr_block = var.cidr

  tags = {
    Name = local.vpc_name
    Env = var.env
  }
}
```

3 CHILD INTERNAL LOGIC

locals.tf

Locals are **not an input interface**. They calculate or standardize values inside the child.

```
locals {
  vpc_name = "${var.vpc_name}-${var.env}"
}
```

Remember:

variable = input received from caller

local = internal calculated value

output = value returned to caller

Same child module, different values:

Dev → 10.0.0.0/16

Test → 10.1.0.0/16

Prod → 10.2.0.0/16



Why variables are declared in BOTH places

The root variable is the root's input. The child variable is the child's input contract. The module block connects them: **cidr = var.vpc_cidr**.

Why use locals?

Locals do not receive values from the root directly. They use child inputs to build reusable internal values, for example a standard name such as **dev-vpc**.

Interview-ready answer

"Root calls the child module and passes values through the module block. The child declares variables to receive those values. The child can then use locals for internal calculations and outputs to return values."